

Random Walk.

$$E_i = \begin{cases} +1 & \text{w.p. } 1/2 \\ -1 & \text{w.p. } 1/2 \end{cases}$$

$$S_0 = a, \quad S_n = \sum_{i=1}^n E_i + a$$

Gambler's Ruin prob

Consider two gamblers A & B with initial capital a & b . Each time a fair coin is tossed, if H turns up, then A gets Rs 1 from B, otherwise B gets Rs. 1 from A.

Total cap $N = a + b$. Let's denote
 the capital of A after n tosses by
 S_n . Then, $S_0 = a$. $S_n = a + \sum_{i=1}^n \epsilon_i$

Cap of B
 $N - S_n$



$P(S_n = N \text{ for some } n \geq 1 \text{ before } S_n = 0 \text{ for some } n \geq 1)$
 $= P(T_N < T_0)$
 $T_i = \text{time to hit } i$

$P(\text{RW returns to the origin starting from the origin}) = 1.$

Basic renewal eqⁿ:

$$f_{2n} = u_{2n-2} - u_{2n}$$

For $n \geq 1$

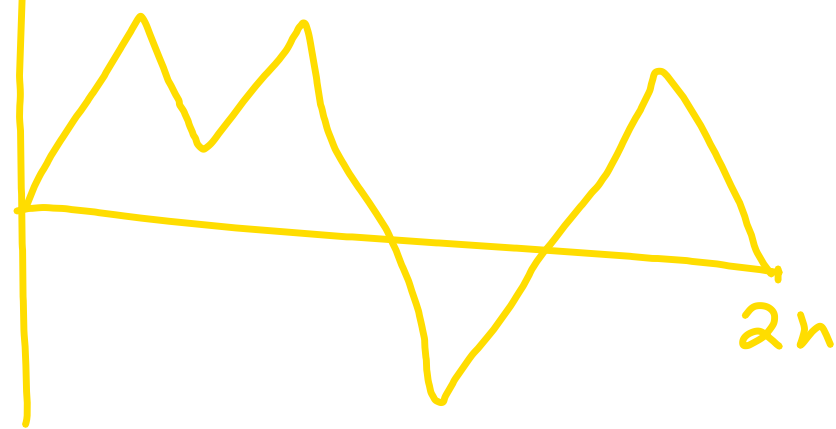
$$\begin{aligned} u_{2n} &= P(S_{2n} = 0) \rightarrow f_{2n} \\ &= P(S_{2n} = 0, S_k \neq 0 \text{ for } 1, 2, \dots, 2n-1) \\ &\quad + P(S_{2n} = 0, S_{2k} = 0 \text{ for some } k=1, \dots, n-1) \end{aligned}$$



$$P(S_{2n}=0, \text{ : } S_{2k} \neq 0 \text{ for some } k=1, \dots, n-1)$$

$$= P(\text{the first return happens}$$

$$\text{at time } 2i, i=1, \dots, n-1)$$



$$B_{2i} = \{S_1 \neq 0, \dots, S_{2i-1} \neq 0, S_{2i} = 0\}$$

$$B_{2i} \cap B_{2j} = \emptyset.$$

$$= P(S_{2n}=0, B_{2i} \text{ happens for some } i=1, \dots, n-1)$$

$$= P(\{S_{2n}=0\} \cap (\bigcup_{i=1}^{n-1} B_{2i}))$$

$$= P\left(\bigcup_{i=1}^{n-1} \left(\{S_{2n}=0\} \cap B_{2i}\right)\right)$$

$$= \sum_{i=1}^{n-1} P\left(B_{2i} \cap \{S_{2n} \neq 0\}\right)$$



$$P(B_{2i} \cap \{S_{2n} \neq 0\})$$

$$= P(S_1 \neq 0, \dots, S_{2i-1} \neq 0, S_{2i} = 0, S_{2n} \neq 0)$$

$$= P(\underbrace{S_1 \neq 0, \dots, S_{2i-1} \neq 0, S_{2i} = 0}_{\text{depends on } (\epsilon_1, \dots, \epsilon_{2i})}, \underbrace{S_{2n} - S_{2i} \neq 0}_{\sum_{k=2i+1}^{2n} \epsilon_k})$$

$$\sum_{k=2i+1}^{2n} \epsilon_k$$

$$= P(S_1 \neq 0, \dots, S_{2i-1} \neq 0, S_{2i} = 0)$$

$$P(S_{2n} - S_{2i} = 0)$$

$$= f_{2i} P(S_{2n-2i} = 0)$$

$$= f_{2i} u_{2n-2i}$$

for $n \geq 1$

$$u_{2n} = f_{2n} + \sum_{i=1}^{n-1} f_{2i} u_{2n-2i}$$

$$= \sum_{i=0}^n f_{2i} u_{2n-2i} = f_{2n} u_0 + \sum_{i=1}^{n-1} \checkmark + u_{2n} f_0.$$

$$\epsilon_1 + \epsilon_2 + \epsilon_3 + \epsilon_4$$

$$\stackrel{d}{=} \epsilon_{2n-3} + \epsilon_{2n-2} + \epsilon_{2n-1} + \epsilon_{2n}$$

$$\sum_{k=2i+1}^{2n} \epsilon_i$$

$$\stackrel{d}{=} \epsilon_1 + \dots + \epsilon_{2n-2i}$$

$$= S_{2n-2i}$$

$\{u_{2n}\}$ is a convolution of $\{f_{2n}\}$ & $\{u_{2n}\}$.
except for $n=0$.

$$F(s) = \sum_{n=0}^{\infty} f_n s^n$$

For $n=0$, $u_0 \neq f_0 u_0 = 0$

relation $\overset{1}{\parallel} \overset{0}{\parallel} \overset{1}{\parallel}$ does not hold.

denote

$$C_n = \sum_{k=0}^n f_k u_{n-k}$$

$C_0 = 0$

$$\begin{aligned} U(s) &= \sum_{n=0}^{\infty} u_{2n} s^{2n} = 1 + \sum_{n=1}^{\infty} u_{2n} s^{2n} \\ &= 1 + U(s) F(s) = 1 + \sum_{n=1}^{\infty} C_{2n} s^{2n} = 1 + \sum_{n=0}^{\infty} C_{2n} s^{2n} \\ &= 1 + \sum_{n=0}^{\infty} C_n s^n \end{aligned}$$

So, $U(s) = 1 + U(s) F(s)$

$$\Rightarrow F(s) = 1 - \frac{1}{U(s)}$$

$$\sum_{n=0}^{\infty} f_n = 1 - \frac{1}{\sum_{n=0}^{\infty} u_n}$$

By Abel's lemma,

$$u_n \sim \frac{1}{\sqrt{\pi n}}$$

$$\sum u_n = \infty$$

$$\sum_{n=0}^{\infty} f_n = \lim_{s \uparrow 1} F(s) = 1 - \frac{1}{\lim_{s \uparrow 1} U(s)} = 1 - \frac{1}{\sum_{n=0}^{\infty} u_n}$$

Thm Prob of no return ^{to origin} until time

$2n$ is u_{2n} .

$$C_k = \{S_1 \neq 0, \dots, S_k \neq 0\}$$

$$P(S_1 \neq 0, S_2 \neq 0, \dots, S_{2n} \neq 0) = u_{2n}.$$

$$\begin{aligned} & P(S_1 \neq 0, \dots, S_{2n-2} \neq 0, S_{2n} = 0) = f_{2n}. \\ & \{S_1 \neq 0, \dots, S_{2n-2} \neq 0\} = C_{2n-2} \cap \{S_{2n-1} \neq 0\} \end{aligned}$$

$$C_{2n-2} = C_{2n-2} \cap \{S_{2n-1} \neq 0\} \rightarrow B_{2n}$$

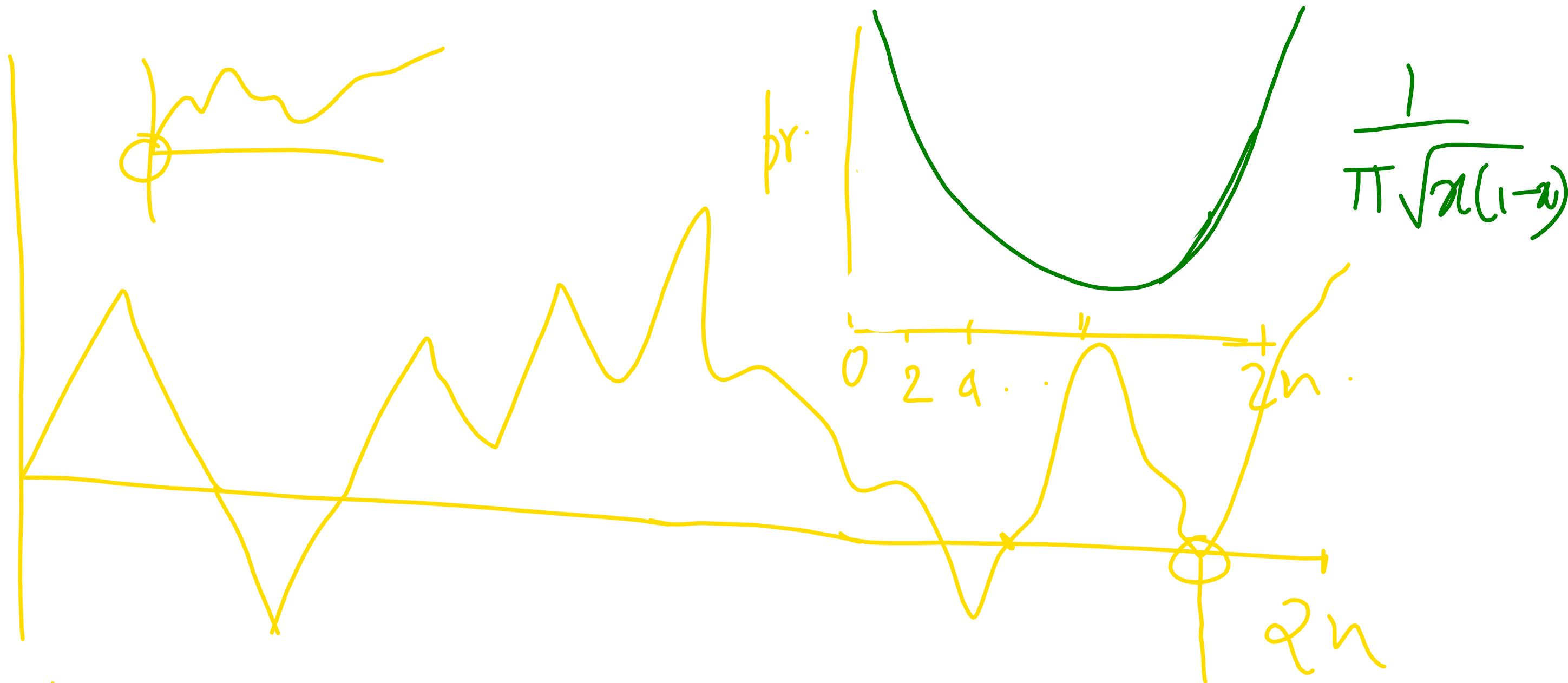
$$= C_{2n-2} \cap \{S_{2n-1} \neq 0, S_{2n} = 0\}$$

$$\cup (C_{2n-2} \cap \{S_{2n-1} \neq 0, S_{2n} \neq 0\}) \rightarrow C_{2n}$$

$$P(C_{2n-2}) = P(B_{2n} \cup C_{2n})$$

$$= P(B_{2n}) + P(C_{2n})$$

$$u_{2n-2} = f_{2n} + u_{2n}.$$



$$W_{2n} = \max \{ k \in \{0, 1, \dots, 2n\} : S_{2k} = 0 \}$$

$$P(W_{2n} = 2k) = U_{2k} U_{2n-2k}.$$

$$\boxed{\frac{W_{2n}}{2n} \Rightarrow W}$$

$W \sim$ are sine
distribⁿ

$$\sim \frac{1}{\sqrt{\pi k}} \cdot \frac{1}{\sqrt{\pi (n-k)}} \\ = \frac{1}{\pi \sqrt{k(n-k)}}$$

$$P(W_{2n} = 2k) = \begin{cases} P(S_1 \neq 0, \dots, S_{2n} \neq 0) & k=0 \\ = u_{2n} \cdot 1 = u_{2n} \cdot u_0 \\ P(S_{2n} = 0) & k=n \\ \text{"} & \\ u_{2n} \cdot u_0 \end{cases}$$

$1 \leq k \leq n-1,$

$$\begin{aligned} P(W_{2n} = 2k) &= P(S_{2k} = 0, S_i \neq 0 \text{ for } i=2k+1, \dots, 2n) \\ &= P(\underbrace{S_{2k} = 0}_{\epsilon_1 \neq \dots \epsilon_{2k}}, \underbrace{S_i - S_{2k} \neq 0 \text{ for } i=2k+1, \dots, 2n}_{\epsilon_{2k+1} \dots \epsilon_{2n}}) \\ &= P(S_{2k} = 0) P(\epsilon_1 \neq \dots \epsilon_{2k}) \\ &= u_{2k} u_{2n-2k}. \end{aligned}$$